

Project Topic: ResuMatch

Problem Statement

Job seekers often struggle to create resumes that are optimized for **Applicant Tracking Systems (ATS)** and properly aligned with specific job descriptions. As a result, qualified candidates may be overlooked during the initial screening process.

There is a need for an accessible and intelligent platform that can analyze resumes, compare them with job descriptions, and provide meaningful feedback to help candidates improve their chances of being shortlisted.

Project Goal

The goal of ResuMatch is to develop a smart, cloud-based resume analysis platform that evaluates resumes and provides an ATS-based matching score. The system helps job seekers understand how well their resume aligns with a particular job description and provides suggestions for improvement.

Introduction

ResuMatch is a smart resume analysis platform designed to help job seekers evaluate and improve their resumes for specific job roles.

The application allows users to upload their resumes and optionally provide a job description. The system processes the resume using AWS cloud services, extracts relevant information, and analyzes the content to generate an ATS-based score, strengths, improvement suggestions, and job recommendations.

The platform is built using a serverless cloud architecture, making it scalable, cost-effective, and easy to maintain.

Features of ResuMatch

1. Cloud-Based Resume Storage

- User-uploaded resumes are stored in **Amazon S3**.
 - Enables secure and scalable cloud storage.
 - Allows resumes to be retrieved for preview, download, and analysis.
-

2. Resume Analysis

The application analyzes uploaded resumes and provides:

- ATS-based matching score
- Resume strengths
- Improvement suggestions
- Tone and language analysis
- Job recommendations

Users can optionally provide a job description to receive a more targeted analysis.

3. Resume History

Users can view their previously uploaded resumes.

The history section provides information such as:

- Resume name
 - Upload date
 - ATS score
 - Resume preview
-

4. Resume Preview, Download, and Delete

The application provides resume management functionality, allowing users to:

- Preview uploaded resumes
 - View analysis reports
 - Download resumes
 - Delete resumes and associated reports
-

5. Serverless Backend Architecture

The backend uses **AWS Lambda** to process application requests without requiring traditional server management.

Lambda functions handle tasks such as:

- Generating presigned S3 URLs
 - Uploading resumes
 - Processing resume analysis requests
 - Fetching resume history
 - Deleting resumes and reports
-

6. Secure API Communication

Amazon API Gateway is used to connect the frontend application with AWS Lambda functions.

It manages API requests for:

- User operations
 - Resume uploads
 - Resume analysis
 - Resume history
 - Resume deletion
-

7. AI-Based Resume Processing

The system uses AWS services to process resume content.

Amazon Textract is used to extract text from uploaded resume documents.

Amazon Comprehend is used to analyze the extracted text and identify relevant information that can assist with resume and job description analysis.

Technology Stack

Frontend

1. HTML5

Used to structure the web pages and application content.

2. CSS3

Used to style the application and create a responsive user interface.

3. JavaScript

Used to add interactivity and communicate with backend APIs.

4. Tailwind CSS

Used to create responsive and modern user interface components.

Backend and Cloud Services

1. AWS Lambda

Used to run backend functions without managing servers.

2. Amazon S3

Used to store uploaded resumes and generated analysis reports.

3. Amazon API Gateway

Used to expose backend APIs and route requests between the frontend and AWS Lambda.

4. Amazon DynamoDB

Used to store and manage application user data.

5. Amazon Textract

Used to extract text from uploaded resume documents.

6. Amazon Comprehend

Used to analyze extracted resume text and identify relevant information for resume analysis.

7. AWS IAM

Used to manage roles and permissions for AWS services and Lambda functions.

8. Amazon CloudWatch

Used for monitoring, logging, and debugging backend Lambda functions and application errors.

9. Amazon CloudFront

Used to deliver the frontend application and static content with improved performance and lower latency.

Project Workflow

Step 1: User Registration and Login

The user creates an account or logs into the ResuMatch application through the frontend interface.

User information is managed through the backend and stored in the application's user database.

Step 2: Resume Upload

The user selects and uploads a resume through the ResuMatch user interface.

A presigned URL is generated by the backend, allowing the resume to be securely uploaded to **Amazon S3**.

Step 3: Job Description Input

The user can optionally provide a job description.

The job description is used to compare the resume with the requirements of a specific job role.

Step 4: Resume Processing and Analysis

After the resume is uploaded:

1. The backend receives the resume analysis request.
 2. The resume is retrieved from Amazon S3.
 3. Amazon Textract extracts text from the resume.
 4. The extracted information is analyzed.
 5. The resume is compared with the provided job description.
 6. An ATS-based score and analysis results are generated.
-

Step 5: Display Analysis Results

The user is presented with the resume analysis results, which may include:

- ATS score
- Strengths and remarks
- Improvement suggestions
- Tone and language analysis
- Job recommendations

The analysis report can also be stored in Amazon S3 for future access.

Step 6: Resume History Management

Users can access their previously uploaded resumes through the **Resume History** section.

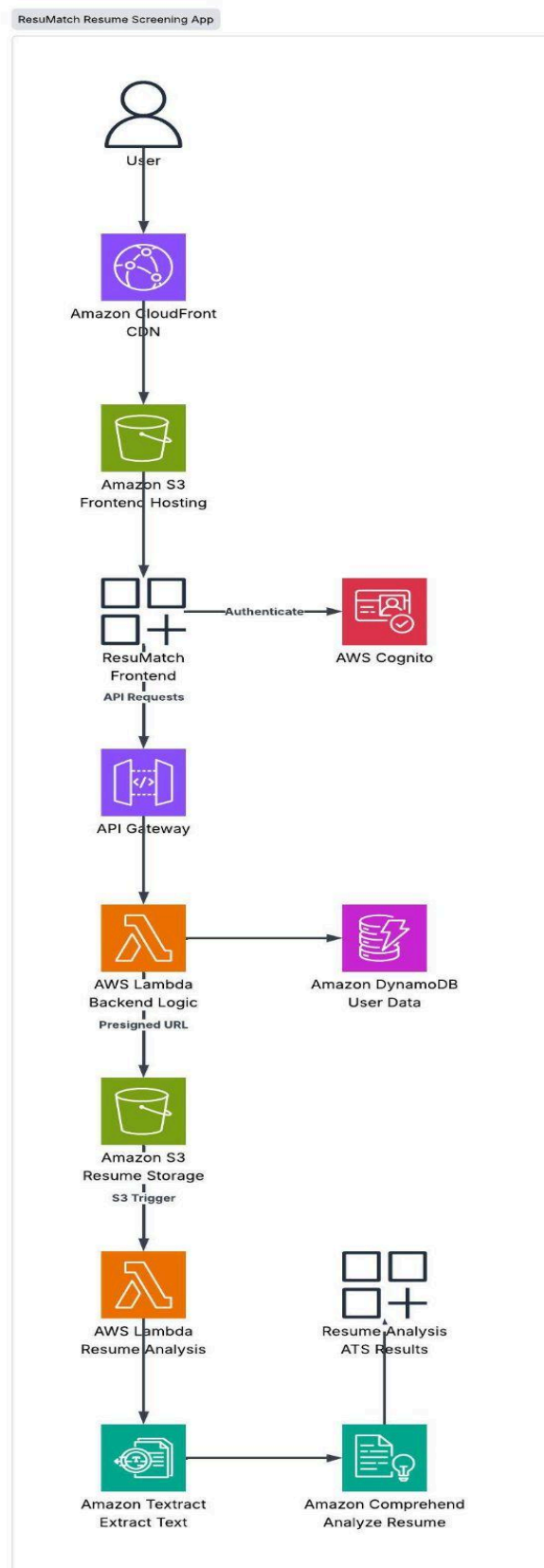
Users can:

- View uploaded resumes
 - Preview resumes
 - Check ATS scores
 - Download resumes
 - Delete resumes
 - Access previous analysis reports
-

Step 7: User Logout

Users can securely log out of the application when they have finished using the platform.

Architectural diagram:



Architecture Diagram Description

The ResuMatch application follows a **serverless and cloud-based architecture** built using AWS services. The overall workflow is as follows:

1. **User:**
The user accesses the ResuMatch application through a web browser.
2. **Amazon CloudFront (CDN):**
Amazon CloudFront delivers the ResuMatch application efficiently to users and acts as a Content Delivery Network (CDN).
3. **Amazon S3 – Frontend Hosting:**
The ResuMatch frontend, including HTML, CSS, and JavaScript files, is hosted in an Amazon S3 bucket and delivered through CloudFront.
4. **ResuMatch Frontend:**
The frontend provides the user interface for authentication, resume uploads, job description input, and displaying the resume analysis results.
5. **AWS Cognito:**
AWS Cognito is used to authenticate and manage users accessing the ResuMatch application.
6. **Amazon API Gateway:**
API Gateway receives requests from the frontend and securely routes them to the backend services.
7. **AWS Lambda – Backend Logic:**
AWS Lambda handles the backend operations, including generating presigned URLs, processing API requests, and managing communication between AWS services.
8. **Amazon DynamoDB – User Data:**
DynamoDB stores user-related information and supports the application's user data management.
9. **Amazon S3 – Resume Storage:**
Uploaded resumes are securely stored in an Amazon S3 bucket using presigned URLs.
10. **AWS Lambda – Resume Analysis:**
When a resume is uploaded, the S3 upload event triggers the Resume Analysis Lambda function.
11. **Amazon Textract:**
Amazon Textract extracts text and relevant information from the uploaded resume.
12. **Amazon Comprehend:**
Amazon Comprehend analyzes the extracted resume text to identify relevant keywords and information required for resume analysis.
13. **Resume Analysis / ATS Results:**
The processed information is used to generate the resume analysis and ATS results, which are then displayed to the user through the ResuMatch frontend.

Overall Flow

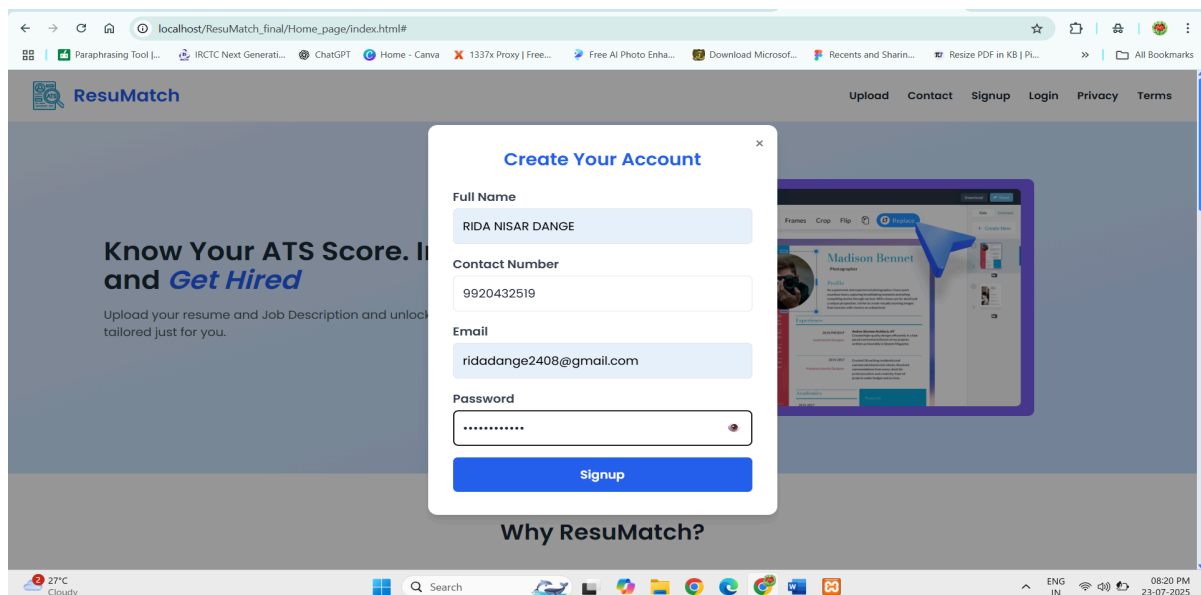
User → CloudFront → S3 Frontend Hosting → ResuMatch Frontend → API Gateway → Lambda → S3 / DynamoDB

For resume analysis:

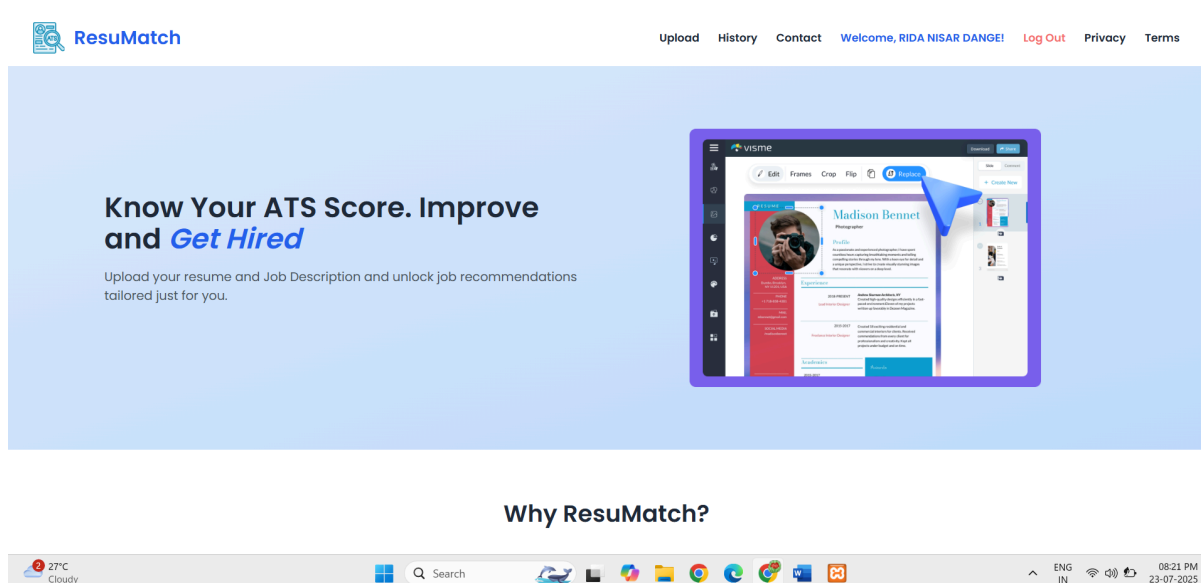
Resume Upload → S3 Resume Storage → Resume Analysis Lambda → Amazon Textract → Amazon Comprehend → Resume Analysis / ATS Results

ResuMatch screenshots:

Signup:



Home page:



Why ResuMatch?

ATS Score

Evaluate how well your resume performs in Applicant Tracking Systems.

Tone & Clarity

Detect tone and improve professional language instantly.

Job Matches

Find matching jobs from your skills and resume keywords.

How ResuMatch Works?

Let *ResuMatch* analyze your resume and match you with the perfect job roles

1

Upload Resume and JD

Upload your resume file or Job Description (if you have any)

2

Get ATS Score

Instantly view your resume's compatibility with ATS systems.

3

Get Job Recommendations

Explore jobs based on your resume and skills.



Analyze

Connect with Me

I'd love to connect! Find me on these platforms:

in q Bē

Ask a Query

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Resume analyze without JD:

Upload Your Resume & Job Description

Upload Resume (PDF/DOCX)

Choose file

UIUX Design resume_Rida.pdf

Paste Job Description (Optional)

Paste the job description here...

Analyze

27°C
Cloudy

Search

ENG
IN

08:22 PM
23-07-2025

Resume analyze with JD:

Upload Your Resume & Job Description

Upload Resume (PDF/DOCX)

Choose File

Application Developer – Cloud Full-Stack Resume.pdf

Paste Job Description (Optional)

Job Description – Cloud Developer

We are looking for a Cloud Developer responsible for designing, developing, deploying, and maintaining cloud-based applications. The candidate should have knowledge of AWS services, serverless architecture, APIs, cloud storage, and security.

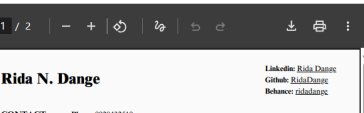
Analyze

Analysis page:


ResuMatch

[History](#)
[Welcome, RIDA NISAR DANGE!](#)
[Log Out](#)

Resume Preview



Resume Analysis Results

ATS Score

Score: 68%

Tone & Language

Your resume tone is *NEUTRAL*.

Strengths / Remarks

- Clear use of action verbs.
- Good structure with relevant sections.

Suggestions

- Include action verbs to highlight achievements.
- Add quantifiable results (e.g., 20% increase in sales).
- Mention relevant certifications to strengthen credibility.

Job Recommendations

- Frontend Developer
- Backend Developer
- Data Scientist
- UI/UX Designer
- Financial Analyst

History page:


ResumeMatch

Your Resume History

Rida N. Dange

Label: Public

Color: Light

Release: Public

CONTACT

Phone: 9024525119

Email: ridadange2023@gmail.com

Address: C. P. Bhabha, Laxdham Complex, Kalyan (E), 421306

CAREER OBJECTIVE

Seeking a challenging career with a progressive organization that provides an opportunity to capitalize my skills and abilities in the field of Information Technology

EDUCATION

1. M.Sc. IT in Ramkrishna Bhambhaniya College (University of Mumbai-CUPB & BT (Grade - A)

Apr21

Uploaded: 2025-07-23

ATS Score: 68%

View

Delete



© 2025 ResumeMatch. Designed and Developed by Rida Dange

27°C

Cloudy

Search



ENG IN

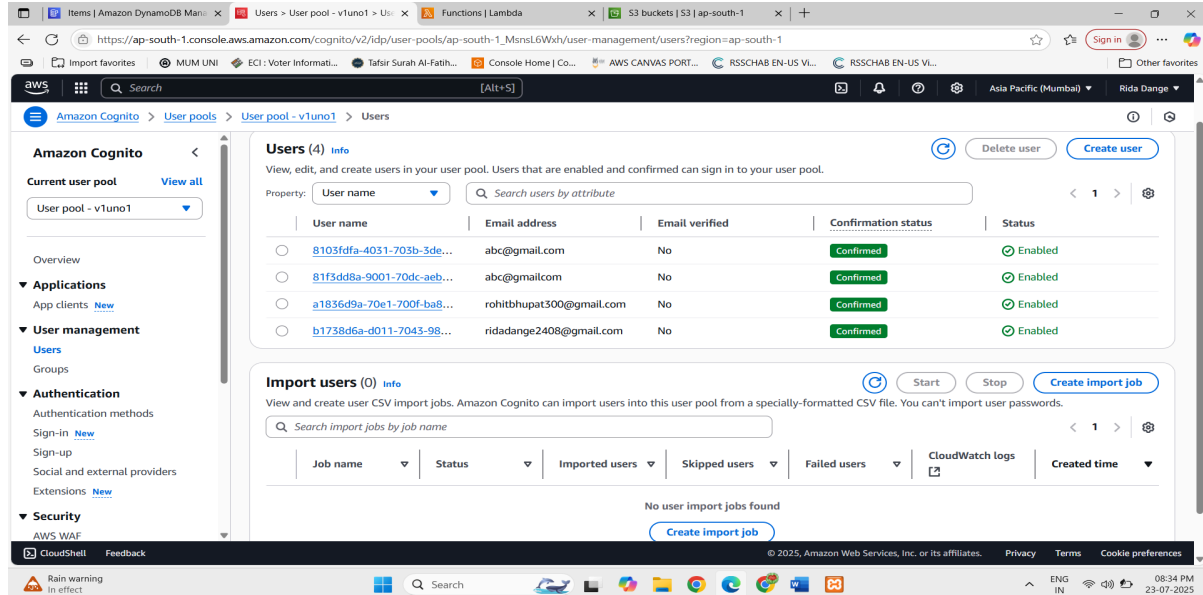


08:24 PM 23-07-2025

AWS Services and Components Used in ResuMatch

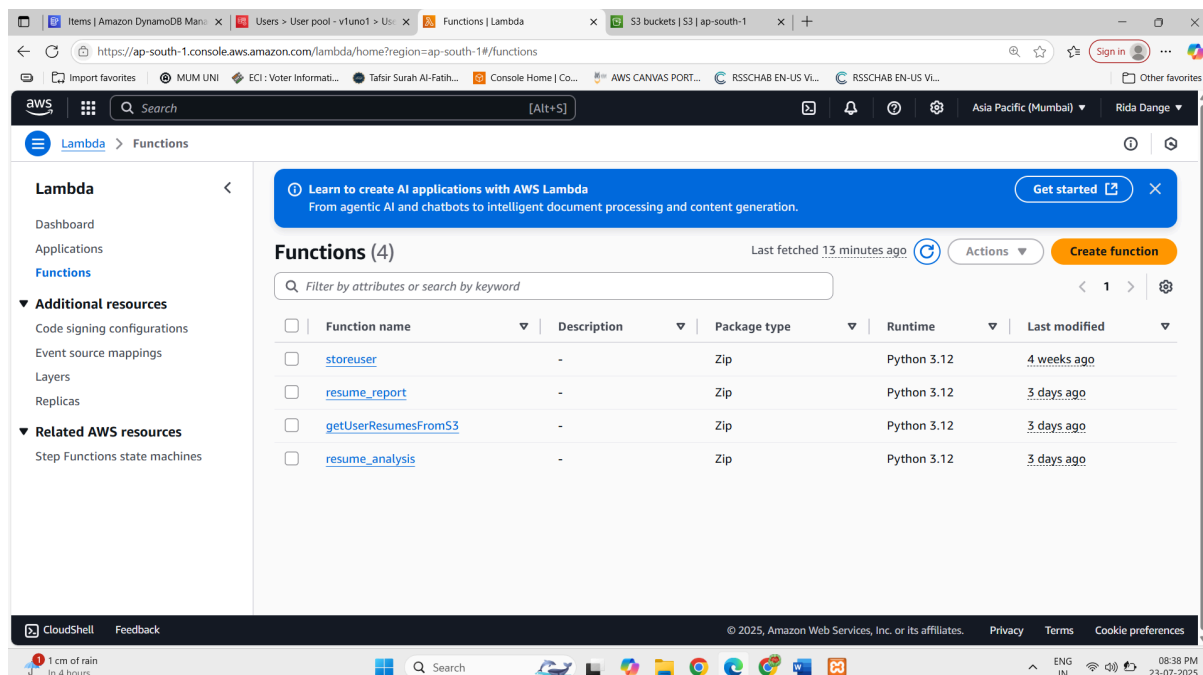
Amazon Cognito

Amazon Cognito is used to provide **secure user authentication and authorization** for the ResuMatch application.



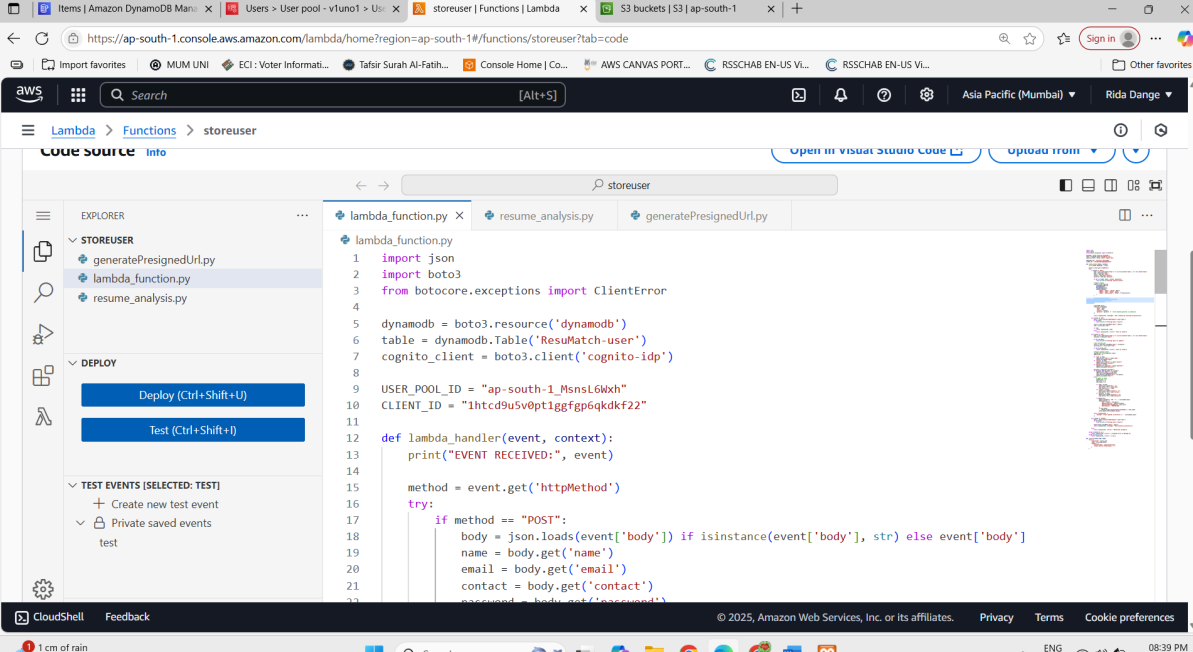
AWS Lambda Functions

AWS Lambda is used to run the backend logic of the ResuMatch application. Different Lambda functions are created to handle different operations.



1. StoreUser Function

The StoreUser Lambda function contains the backend logic for storing user information. It processes user registration data received from the frontend and stores the relevant user details in Amazon DynamoDB.

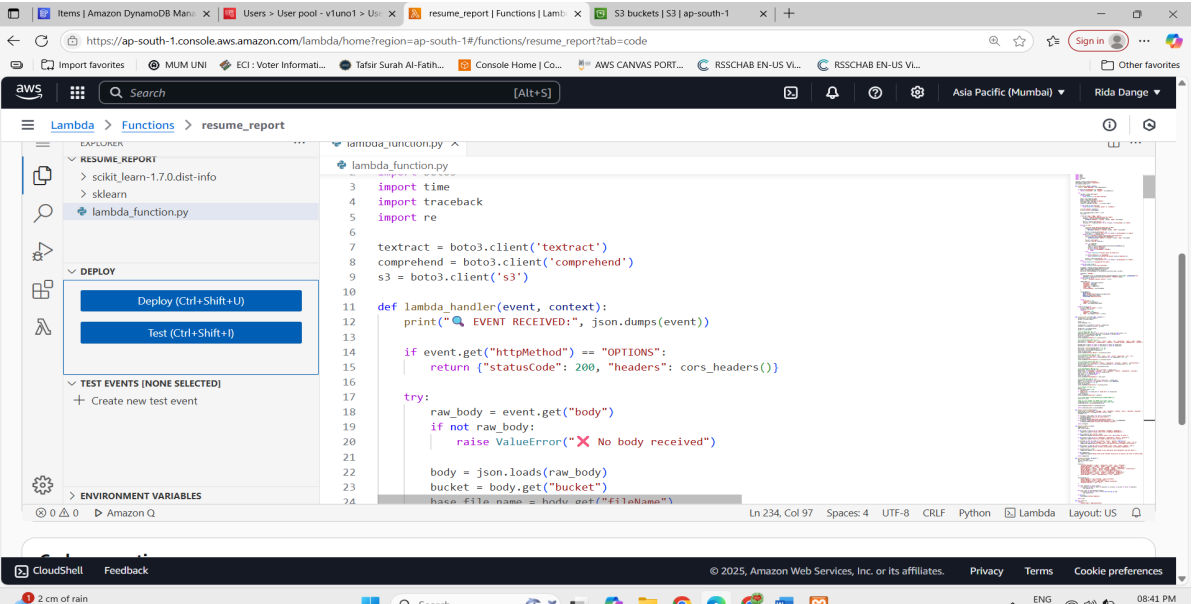


The screenshot shows the AWS Lambda console for the 'storeuser' function. The code is as follows:

```
1 import json
2 import boto3
3 from botocore.exceptions import ClientError
4
5 dynamodb = boto3.resource('dynamodb')
6 table = dynamodb.Table('ResumeMatch-user')
7 cognito_client = boto3.client('cognito-idp')
8
9 USER_POOL_ID = "ap-south-1_MsnsL6wkh"
10 CLIENT_ID = "1htcd9u5v0pt1ggf6p6qdkf22"
11
12 def lambda_handler(event, context):
13     print("EVENT RECEIVED:", event)
14
15     method = event.get('httpMethod')
16     try:
17         if method == "POST":
18             body = json.loads(event['body']) if isinstance(event['body'], str) else event['body']
19             name = body.get('name')
20             email = body.get('email')
21             contact = body.get('contact')
22             password = body.get('password')
```

2. Resume Report Function

The Resume Report Lambda function handles the generation and storage of resume analysis reports. After a resume is analyzed, the generated report, including the ATS score and other analysis results, is processed and stored for future access.



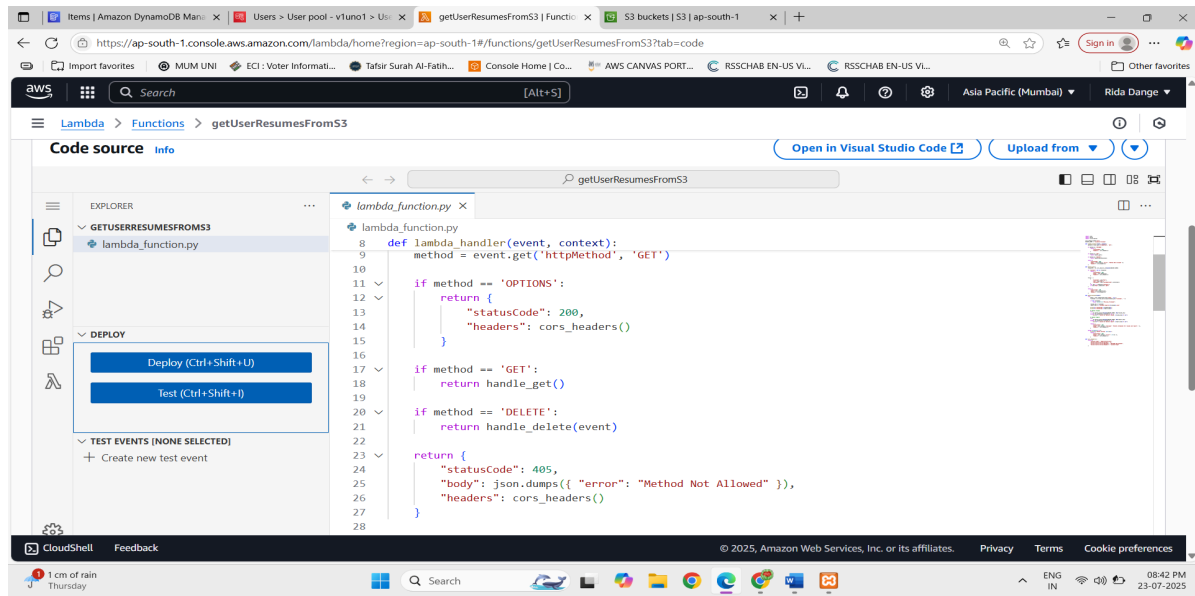
The screenshot shows the AWS Lambda console for the 'resume_report' function. The code is as follows:

```
3 import time
4 import traceback
5 import re
6
7 textract = boto3.client('textract')
8 comprehend = boto3.client('comprehend')
9 s3 = boto3.client('s3')
10
11 def lambda_handler(event, context):
12     print("EVENT RECEIVED:", json.dumps(event))
13
14     if event.get("httpMethod") == "OPTIONS":
15         return {"statusCode": 200, "headers": cors_headers()}
16
17     try:
18         raw_body = event.get("body")
19         if not raw_body:
20             raise ValueError("No body received")
21
22         body = json.loads(raw_body)
23         bucket = body.get("bucket")
24         file_name = body.get("fileName")
```

3. getUserResumesFromS3 Function

The **getUserResumesFromS3** Lambda function is responsible for retrieving previously uploaded resumes from the Amazon S3 bucket.

This function fetches the available resume files from the backend and sends the information to the frontend, where the resumes are displayed on the **Resume History** page. It also handles resume deletion requests from the application.

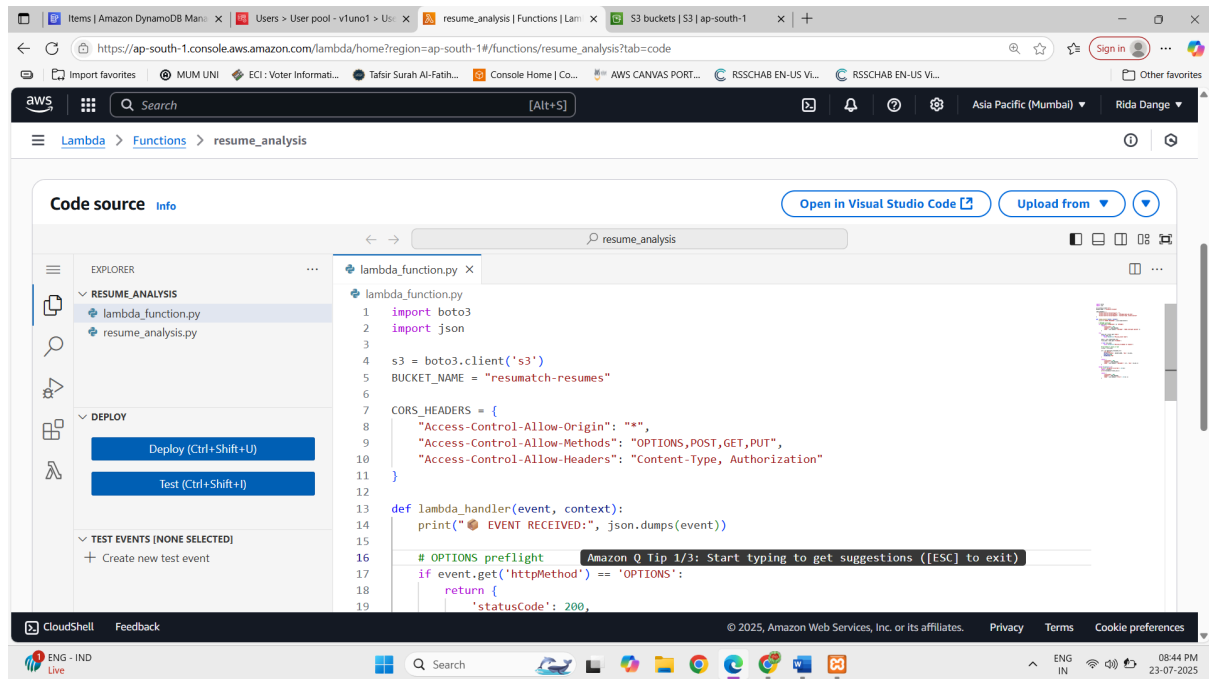


4. Resume Analysis Function

The **Resume Analysis** Lambda function contains the main backend logic for analyzing uploaded resumes.

The function processes the resume stored in Amazon S3, extracts and analyzes the resume content, compares it with the provided job description (if available), and generates results such as:

- ATS score
- Resume strengths
- Improvement suggestions
- Tone and language analysis
- Job recommendations



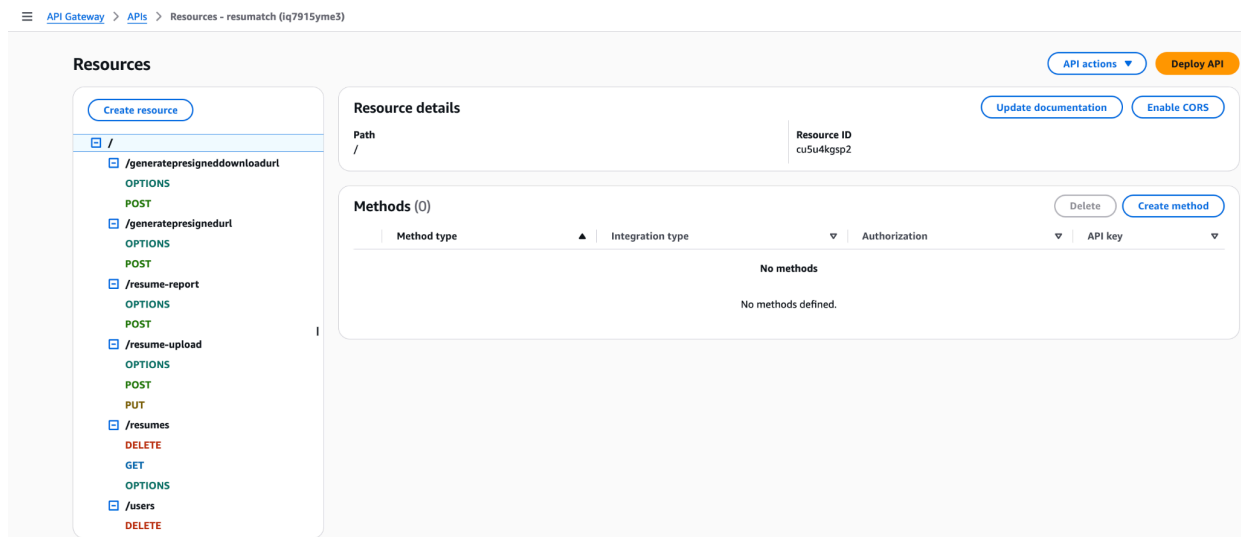
Amazon API Gateway

Amazon API Gateway acts as a bridge between the **ResuMatch frontend and backend**. It receives requests from the frontend and routes them to the appropriate AWS Lambda functions.

It is used for:

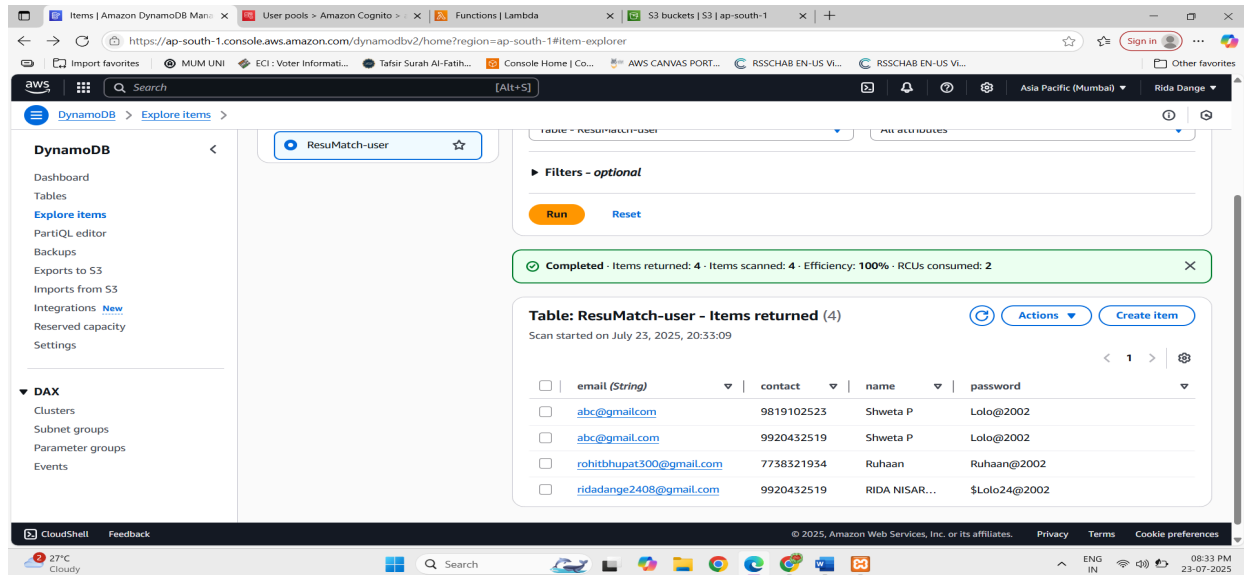
- User registration and login
- Resume upload and analysis
- Fetching resume history
- Deleting resumes

Flow: Frontend → API Gateway → AWS Lambda → AWS Services



Amazon DynamoDB

Amazon DynamoDB is used to store and manage user-related data for the ResuMatch application. This includes user information such as registration and login details.



Amazon S3 Bucket

Amazon S3 is used as the cloud storage service for the ResuMatch application.

The S3 bucket stores:

- Resumes uploaded by users from the frontend
- Resume analysis reports
- Frontend source code and static website files

S3 provides scalable and reliable cloud storage for the application.

Items | Amazon DynamoDB Man... | Users > User pool - v1tuno1 > Us... | resume_analysis | Functions | Lam... | resumatch-resumes - S3 bucket |

https://ap-south-1.console.aws.amazon.com/s3/buckets/resumatch-resumes?region=ap-south-1&bucketType=general

Sign in

Other favorites

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[Alt+S]

Asia Pacific (Mumbai) | Rida Dange

Amazon S3

General purpose buckets

Directory buckets

Table buckets

Vector buckets [Preview](#)

Access Grants

Access Points (General Purpose Buckets, FSx file systems)

Access Points (Directory Buckets)

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens

Dashboard

CloudShell

Feedback

resumatch-resumes [Info](#)

Objects | Properties | Permissions | Metrics | Management | Access Points

Objects (1/2)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

	Name	Type	Last modified	Size	Storage class
☐	resumes-reports/	Folder	-	-	-
☒	UI/UX Design resume_Rida.pdf	pdf	July 23, 2025, 20:23:10 (UTC+05:30)	80.3 KB	Standard

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Sports headline Manchester Unit...

Search

ENG IN 08:45 PM 23-07-2025

Amazon S3 > Buckets > resumatch-frontend-new

resumatch-frontend-new [Info](#)

Objects | Metadata | Properties | Permissions | Metrics | Management | File systems - new | Access Points

Objects (9)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

	Name	Type	Last modified	Size	Storage class
☐	auth.js	js	August 27, 2026, 09:47:44 (UTC+05:30)	2.0 KB	Standard
☐	Dashboard/	Folder	-	-	-
☐	History_page/	Folder	-	-	-
☐	Home_page/	Folder	-	-	-
☐	index.html	html	August 27, 2026, 11:05:19 (UTC+05:30)	17.3 KB	Standard
☐	Privacy&Terms/	Folder	-	-	-
☐	ResuMatch project doc.docx	docx	August 24, 2026, 21:57:06 (UTC+05:30)	3.7 MB	Standard
☐	Resume Analysis/	Folder	-	-	-
☐	resume-analysis/	Folder	-	-	-

Amazon CloudFront

Amazon CloudFront is used as a **Content Delivery Network (CDN)** to deliver the ResuMatch frontend application to users.

The frontend source code is stored in the Amazon S3 bucket, and CloudFront distributes the website content efficiently to users.

CloudFront helps improve:

- Website loading speed
- Performance
- Content delivery
- Scalability

The overall frontend delivery flow is:

User → CloudFront → Amazon S3

CloudFront > Distributions > E2OMBUKYCV1ANM

resumatch Free plan View metrics

Details

Distribution domain name

 d1xf95v64u2r03.cloudfront.net

Billing

Free plan (\$0/month)

Manage plan

ARN

 arn:aws:cloudfront::340705233280:distribution/E2OMBUKYCV1ANM

Last modified

August 26, 2026 at 9:44:14 AM UTC

General

Security

Origins

Behaviors

Error pages

Invalidations

Logging

Tags

Settings

Name

resumatch 

Description

-

Price class

Use all edge locations (best performance)

Supported HTTP versions

HTTP/2, HTTP/1.1, HTTP/1.0

Standard logging

Available with the Pro plan

Default root object

-

Edit